

JEE-Main-29-01-2024 (Memory Based)

[MORNING SHIFT]

Physics

Question: A block of mass 100 kg is moved along a horizontal surface 10 m from the starting point. If coefficient of friction between ground and the block is 0.4 find work done against friction

Options:

- (a) 3.9 kJ
- (b) 4.2 kJ
- (c) 3.7 kJ
- (d) 4.1 kJ

Answer: (a)

Question: A particle is executing SHM with an amplitude A. If potential energy of the system is zero about mean position $x = 0$, Find ratio of total energy to kinetic energy at $x = A/3$

Options:

- (a) 8/9
- (b) 9/8
- (c) $3/2\sqrt{2}$
- (d) $2\sqrt{2/3}$

Answer: (b)

Question: $i = 20 + 3/2 t$ Find charge flown in 20 S

Options:

- (a) 1600 C
- (b) 1200 C
- (c) 1000 C
- (d) 800 C

Answer: (c)

Question: Match the following

A	$\oint \underline{B} \cdot d\underline{A} = 0$	P	Faraday & Lenz's law
B	$\oint \underline{E} \cdot d\underline{A} = \frac{Q_{in}}{\epsilon_0}$	Q	Gauss law on magnetism
C	$\oint \underline{B} \cdot d\underline{l} = \mu_0 i_{enc}$	R	Ampere's law

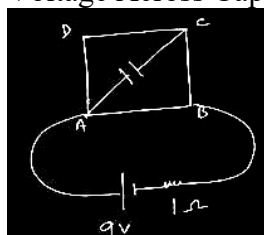
D	$\oint \underline{E} \cdot d\underline{l} = -\frac{d\phi_B}{dt}$	S	Gauss law of electrostatics
---	--	---	-----------------------------

Options:

- (a) (A-Q), (B-S), (C-R), (D-P)
- (b) (A-S), (B-Q), (C-R), (D-P)
- (c) (A-Q), (B-R), (C-S), (D-P)
- (d) (A-Q), (B-S), (C-P), (D-R)

Answer: (a)

Question: In the Following Circuit the resistance of square loop ABCD is 16 Ohm. Find the Voltage Across Capacitor in steady State



Options:

- (a) 4.5 V
- (b) 4 V
- (c) 3 V
- (d) 1 V

Answer: (a)

Question: A Square loop of side 0.1 m is in East West Plane and magnetic field is along North East of 0.2 T. If B is Removed in 10 s find EMF Induced?

Options:

- (a) 14 mV
- (b) 0.14 mV
- (c) 1 mV
- (d) 0.2 mV

Answer: (b)

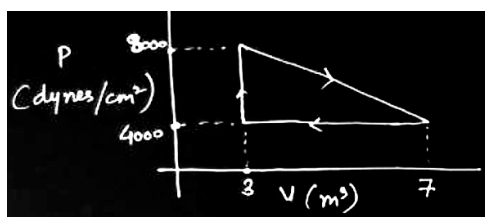
Question: If de Broglie wavelength of an electron is same as wavelength of a photon and speed of the electron is 25% of speed of EM waves in vacuum. Find ratio of kinetic energy of electron & energy of photon.

Options:

- (a) 1/8
- (b) 1/4
- (c) 1/2
- (d) 1

Answer: (a)

Question: P-V graph of a gas is given. Find the work done by the gas.



Options:

- (a) 400 J
- (b) 600 J
- (c) 800 J
- (d) 100 J

Answer: (c)

Question: A convex lens made of glass ($\mu_{\text{glass}} = 1.5$) has focal length of 20 cm in air. If this lens is put inside a fluid of refractive index 1.6. The new focal length will be

Options:

- (a) 160 cm
- (b) -160 cm
- (c) -180 cm
- (d) 80 cm

Answer: (b)

Question: If R is the radius of Earth's and Particle has Equal weight at " d " distance below the surface of Earth's and " d " distance above it, find " d "

Options:

- (a) $d = \sqrt{5} R/2$
- (b) $d = \sqrt{3} R$
- (c) $d = (\sqrt{5}-1) R/2$
- (d) $d = R$

Answer: (c)

Question: The flow speeds on upper & lower surfaces of the wings are 70 m/s & 64 m/s respectively on an airplane in a wind tunnel. What is the lift force on the wing? Area of wing is 0.2 m^2 . Given: density of air = 1.2 kg/m^3

Options:

- (a) 16
- (b) 36
- (c) 81
- (d) 144

Answer: (c)

Question: In a concave mirror of radius of curvature $R = 30 \text{ cm}$ the size of inverted image is half the size of object. Find the distance of the object from pole.

Options:

- (a) 30
- (b) 45
- (c) 60
- (d) 20

Answer: (b)

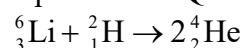
Question: A Galvanometer shows deflection corresponding to 25 division when a certain current is passed. The deflection becomes 5 divisions when galvanometer is shunted with 24Ω . Find the resistance of galvanometer

Options:

- (a) 24Ω
- (b) 48Ω
- (c) 96Ω
- (d) 120Ω

Answer: (c)

Question: In the given nuclear reaction, which of the following expression correctly represent the Q value



Given masses:

$${}^6_3\text{Li} = 6.015122 \text{ amu}, {}^4_2\text{He} = 4.002603 \text{ amu}$$

$${}^2_1\text{H} = 2.014101 \text{ amu}, 1 \text{ amu} = 931.5 \text{ MeV}$$

Options:

- (a) 22.37 MeV
- (b) 21.42 MeV
- (c) 22.02 MeV
- (d) 21.90 MeV

Answer: (a)

Question: S1: When a capillary tube is dipped in cold water and then hot water, the height of water increases

S2: When a capillary tube is dipped in hot water and then cold water, the height of water decreases

[Assume negligible change in density of water or radius of capillary]

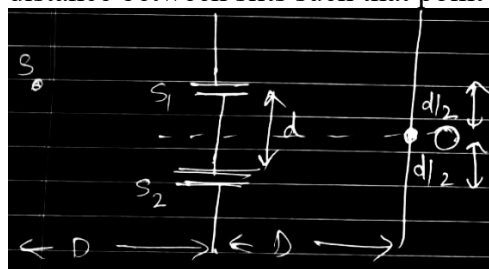
Options:

- (a) 1 true, 2 false
- (b) 1 false, 2 true
- (c) Both false
- (d) Both True

Answer: (c)

Question: In YDSE experiment source is placed exactly in front of one slit.

The distance between slits & screen is 0.2m. Wavelength used is 400 nm. Find the minimum distance between slits such that point O is dark



Options:

- (a) 0.28 mm
- (b) 0.36 mm

- (c) 0.14 mm
(d) 0.49 mm

Answer: (a)

Question: A galvanometer with resistance $R_g = 8\Omega$ has a full scale deflection current of $I_g = 3 \text{ mA}$. What is the shunt resistance required to create an ammeter of 8 ampere range?

Options:

- (a) 0.001Ω
(b) 0.003Ω
(c) 0.009Ω
(d) 0.01Ω

Answer: (b)

Question: Calculate the flux passing through a sphere of radius $4a$ whose center is at the origin, if two charges $5q$ and $-2q$ are placed at $(2q, 0)$ and $(-5q, 0)$ respectively

Options:

- (a) $5q/\epsilon_0$
(b) $-2q/\epsilon_0$
(c) $7q/\epsilon_0$
(d) $3q/\epsilon_0$

Answer: (a)

Question: If the magnetic potential due to a small magnetic dipole along the axis at a distance of 20 cm is $1.5 \times 10^{-5} \text{ J Am}^{-1}$ find its magnetic dipole moment

Options:

- (a) 4 Am^2
(b) 6 Am^2
(c) 8 Am^2
(d) 2 A^2

Answer: (b)

Question: A solid cylinder rolls down an incline plane of 60° with acceleration of $x/\sqrt{2}$. Find x .

Options:

- (a) 10
(b)
(c)
(d)

Answer: (a)

Question: $U = 2x^2 + 3y^3 + 2z$ Find X component of force in Newtons at $x = 1 \text{ m}$?

Options:

- (a) 4
(b) -4
(c) 2
(d) -2

Answer: (b)

Question: Ratio of centripetal acceleration of two particles in a circles of same radii is 3 : 4.
Then the ratio of their tangential speed is ?

Options:

(a) 3 : 2

(b) $\sqrt{3} : 2$

(c) $\sqrt{3} : 4$

(d) $\sqrt{3} : \sqrt{2}$

Answer: (b)



JEE-Main-29-01-2024 (Memory Based)

[MORNING SHIFT]

Chemistry

Question: Which of the following pair will be formed by the decomposition of KMnO_4 ?

Options:

- (a) MnO_4 , MnO_2
- (b) K_2MnO_4 , MnO_2
- (c) KMnO_4 , MnO_2
- (d) MnO_2 , H_2O

Answer: (b) K_2MnO_4 , MnO_2

Solution:

Potassium permanganate forms dark purple (almost black) crystals which are isostructural with those of KClO_4 . The salt is not very soluble in water (6.4 g / 100 g of water at 293 K), but when heated it decomposes at 513 K.



Question: Interaction b/w π . Bond & lone pair l-s on adjacent atoms

Options:

- (a) Resonance
- (b) Hyper conjugation
- (c) Inductive Effect
- (d) Electronic Effect

Answer: (a) Resonance

Solution:

Question: Assertion. Electronegativity increase across a period

Reason. Effective increase in nuclear charge is more than effective shielding.

Options:

- (a) Step 1: Electronegativity increase down the group 14 is to pb
- (b) Step 2: Group 14 contains metals, non metals and also metalloids

Solution: Assertion true reason true

Step : 1 is incorrect but Step : 2 is correct

Question:

Column - I	Column - II
Ziegler Natta Catalyst	Rh
Blood Pigment	CO
Wilkinson Catalyst	Fe
Vitamin B12	Ti

Solution:

- 1 → Ti
 2 → Fe
 3 → Rh
 4 → Co

Question: Appearance of Red colour on treatment with Na fusion extract of an organic compound with FeSO_4 in presence of conc. H_2SO_4 indicate element

Options:

- (a) N
 (b) Br
 (c) S
 (d) N & S

Answer: (d) N & S

Solution: (d) N & S

Question: Cl^- shows disproportionation in alkaline meol :

**Options:**

- (a) 1 1 1 3
 (b) 3 6 2 4
 (c) 1 2 1 1
 (d) 2 4 1 3

Answer: (b)

Solution: $3\text{Cl}_2 + 6\text{OH}^- \rightarrow 2\text{ClO}_3^- + 4\text{Cl}^- + 3\text{H}_2\text{O}$

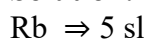
Question: The correct set of 4 Quantum numbers of Valence e^- of Rb(37)

Options:

- (a) $n = 5$., $l = 0$., $m = 1$.,
 (b) $n = 5$., $l = 0$., $m = 0$.,
 (c) $n = 5$., $l = 1$., $m = 0$.,
 (d) $n = 5$., $l = 1$., $m = 1$.,

Answer:

Solution:



↓

$$n = 5$$

$$l = 0$$

$$Ml = 0$$

$$Mg = +1/2 \text{ or } -1/2$$

The electronic configuration of rubidium atom ($Z = 37$) is given by



Hence, the quantum numbers for $5s^1$ electron is given by

$$n = 5, l = 0, m = 0, s = +1/2 \text{ or } -1/2$$

Question: Type of amino acids obtained on hydrolysis of proteins

Options:

- (a) α
 (b) β

(c) γ

(d) δ

Answer: (a)

Solution: Alpha amino acid

Question: CO forms a bridge b/w M atoms

Options:

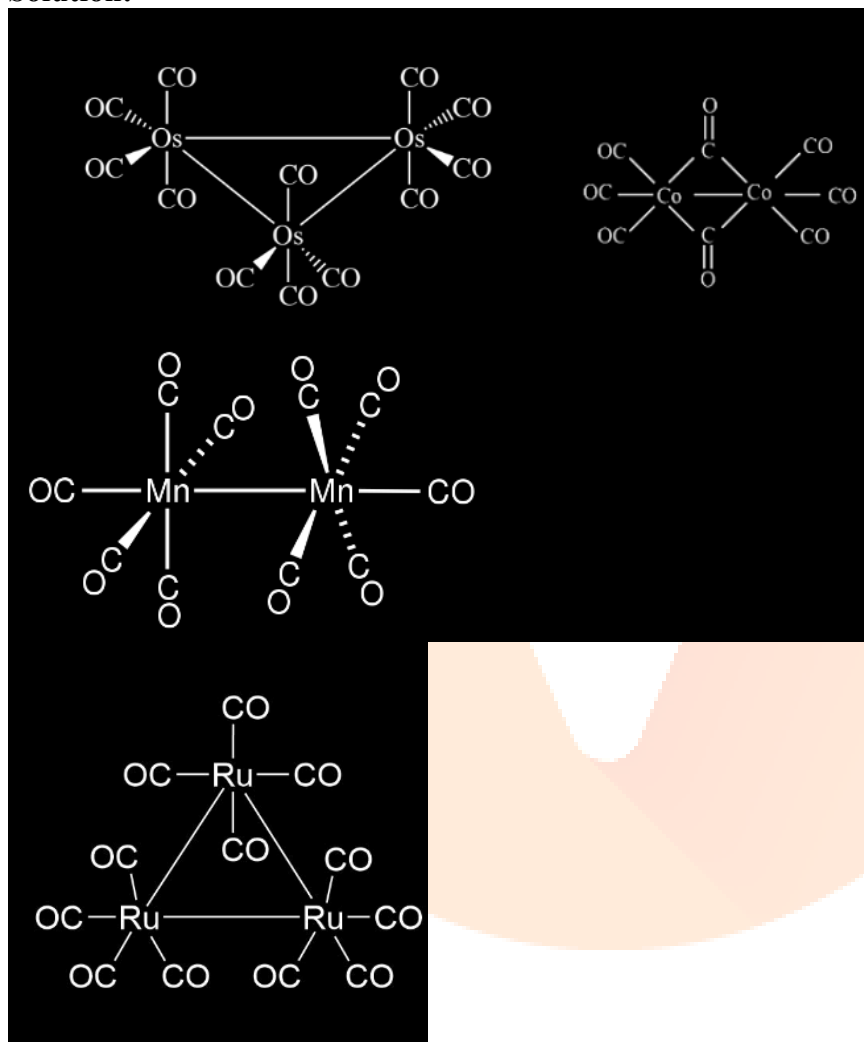
(a) $\text{Os}_3(\text{CO})_{12}$

(b) $\text{Co}_2(\text{CO})_8$

(c) $\text{Ru}_3(\text{CO})_{12}$

(d) $\text{Mn}_2(\text{CO})_{10}$

Solution:



Question: Calculate the Molarity of a Solution having density = 1.25 g/ml. % (w/w) of Solute is 31.4% of H_2SO_4 solution

Options:

(a) 4

(b) 9

(c) 8

(d) 6

Answer: (a)

Solution:

$$M = 10 \times w/w \% \times d$$

$$M = 10 \times \frac{M_{\text{solute}}}{31.4} \times \frac{125}{98} \times 100$$

$$= 4$$

Question: Find all quantum numbers $Z = 37$

Options:

- (a) $n = 5$., $l = 0$., $m = 1$.,
- (b) $n = 5$., $l = 0$., $m = 0$.,
- (c) $n = 5$., $l = 1$., $m = 0$.,
- (d) $n = 5$., $l = 1$., $m = 1$.,

Answer: (a)

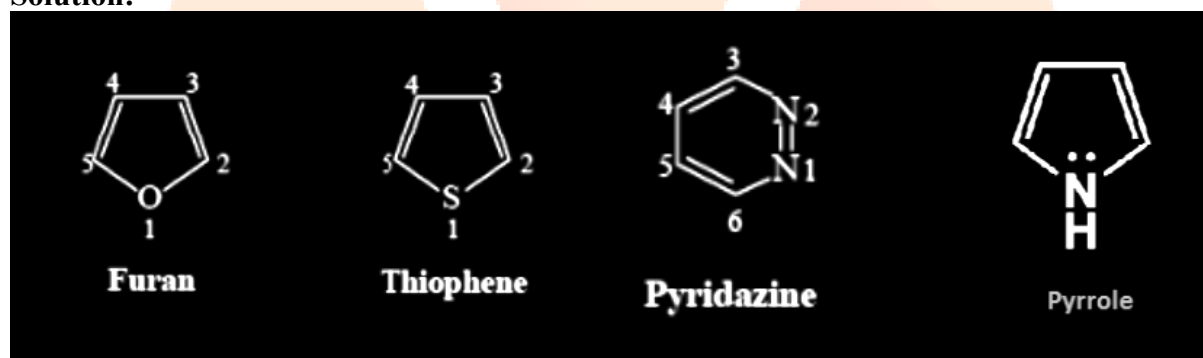
Question: Among the heterocyclic compound that contain Sulphur atom is :

Options:

- (a) Pyridazine
- (b) Furan
- (c) Thiophene
- (d) Pyrrole

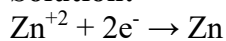
Answer: (c)

Solution:



Question: Find weight of Zinc in Zinc sulphate electrolysis $i = 0.015 \text{ A}$ $t = 15 \text{ minutes}$

Solution:



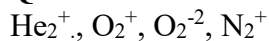
$$1 \text{ mol Zn} = 65.3 \text{ gm} = 2 \text{ F}$$

$$\text{Number of Faradays} = \frac{0.015 \times 15 \times 60}{965}$$

$$= 0.00013 \text{ g F}$$

$$= .0046$$

Question: Number of compound in which B.O = 1 and is paramagnetic



Answer: 0

Solution:

	B.O	Magnetic nature
He_2^+	0.5	Paramagnetic
O_2^+	1.5	Paramagnetic
O_2^{-2}	1	Diamagnetic

N_2^+

2.5

Paramagnetic

Question: Number of compounds that gives positive fehling test Benzaldehyde, acetophenone, methanal

Answer: 1

Solution: Aliphatic aldehyde group. Aromatic aldehydes and ketones do not a give Fehling's test.



JEE-Main-29-01-2024 (Memory Based)

[MORNING SHIFT]

Maths

Question: $f(x) = 2^x - x^2$ m = number of solution such that $f(x)$ with x axis
 N = number of solutions such that $f'(x)$ with x axis $m + n$?

Answer: 5

Solution:

$$f(x) = 2^x - x^2$$

$$f'(x) = 2^x \ln 2 - 2x$$

$$m = 3$$

$$n = 2$$

Question: $(1 + y^2)(1 + \ln x) dx + x dy = 0$

Answer:

Question: Find $I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{x^2 \cos x}{1 + e^x} + \frac{1 + \sin^2 x}{1 + e^{\sin(x^{2023})}} \right) dx$

Solution:

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x^2 \sin x}{1 + e^x} + \frac{1 + \cos^2 x}{1 + e^{\sin(x^{2023})}} dx$$

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x^2 \sin x}{1 + e^x} - e^x + \frac{1 + \cos^2 x}{1 + e^{\sin(x^{2023})}} dx$$

$$\alpha I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x^2 \sin x + 1 + \cos^2 x$$

$$\cancel{\alpha} I = \cancel{\alpha} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x^2 \sin x + \cos^2 x dx$$

$$I = x^2 (-\cos x) + (2x)(+\sin x) + 2(\cos x) \Big|_0^{\frac{\pi}{2}} + \frac{\pi}{2} + \frac{\pi}{4}$$

$$I = \left(0 + \cancel{2} \cdot \frac{\pi}{2} \right) - (0 + 0 + 2) + \frac{3\pi}{4}$$

$$I = \frac{7\pi}{4} - 2$$

Question: If an AP with terms $\langle a_i \rangle$, $a_6 = 2$ and a_1, a_4, a_5 is maximum. Find the common difference.

Solution:

$$a_6 = 2 \text{ \& } a_1 a_4 a_5 = \max. (\text{given})$$

$$\Rightarrow M = (a_6 - 5d)(a_6 - 2d)(a_6 - d)$$

$$= (2 - 5d)(2 - 2d)(2 - d)$$

$$M = 2(-5d^3 + 17d^2 - 16d + 4)$$

$$\frac{dM}{dQ} = 2(-15d^2 + 34d - 16) = 0$$

$$= 2(-3d(5d - 8) + 2(sd - 8)) = 0$$

$$= -2(3d - 2)(5d - 8) = 0$$

$$d = \frac{8}{5}$$

Question: $\frac{{}^{11}C_1}{2} + \frac{{}^{11}C_2}{3} + \dots + \frac{{}^{11}C_9}{9} = \frac{n}{m}$

gcd (M, n) = 1, find m + n.

Solution:

$$(1+x)^{11} = \sum_{r=0}^{r=11} {}^h C_r x^r$$

$$\int_0^1 (1+x)^{11} dx = \sum_{r=0}^{r=11} \frac{{}^{11}C_r x^{r+1}}{r+1} \Big|_0^1$$

$$\frac{2^{12}-1}{12} = \sum_{r=0}^{r=11} \frac{{}^{11}C_r}{r+1}$$

$$= \frac{{}^{11}C_0}{1} + (5) + \frac{{}^{11}C_8}{10} + \frac{{}^{11}C_{10}}{11} + \frac{{}^{11}C_{11}}{12}$$

$$\frac{2^{12}-1}{12} = 5 + \frac{91}{12} \Rightarrow 5 = \frac{4096-91}{12} = \frac{4095}{12} = \frac{1365}{4}$$

$$m+n=1369$$

Question: $f(x) = \frac{(2^x + 2^{-x})(\tan x) \sqrt{\tan^{-1}(x^2 - x + 1)}}{(x^3 - x^2 + 1)^3}$

Find $f'(a) =$

Solution:

$$f(x) = \frac{(2^x + 2^{-x}) \tan x \sqrt{\tan^{-1}(x^2 - x + 1)}}{(x^3 - x^2 + 1)^3}$$

$$f(0) = 0$$

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x)}{x} = \frac{2 \cdot 1 \cdot \sqrt{\frac{\pi}{4}}}{1} = \sqrt{\pi}$$

Question:

$$\left. \begin{aligned} x^2 + y^2 &= 46, \\ \frac{x^2}{16} + \frac{y^2}{b^2} &= 1 \end{aligned} \right\}$$

POI lies on $y^2 = 3x^2$ find $3\sqrt{3}$ times of areas of rectangle formed by POI of conics

Question:

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{1 \cdot \int_{x^3}^{\left(\frac{\pi}{2}\right)^3} \cos\left(x^{\frac{1}{3}}\right) dx}{\left(x - \frac{\pi}{2}\right)^2}$$

Solution:

$$= \frac{3\pi^2}{8}$$

Question: GTWENTY, find rank of GTWENTY

Solution:

2 4 5 1 3 4 6

G T W N T Y

$\frac{1}{2}! \quad \frac{2}{2}! \quad 3 \quad 0 \quad 0 \quad 0 \quad 0$

$$\text{Rank} = \frac{1}{2!} \times 6! + \frac{2}{2!} \times 5! + 3 \times 4! + 1$$

$$= 360 + 120 + 72 + 1$$

$$= 480 + 73 = 553$$

Question:

$$A \cdot A^T = I \text{ Value of } \left(\frac{1}{2}A\right) \left[(A + A^T)^2 + (A - A^T)^2 \right]$$

Options:

(a) $A^3 + AT$

(b) $(A^3 + AT)^2$

(c) $(A^3 + I)$

(d) A^3

Solution:

$$A \cdot A^T = I$$

$$\frac{1}{2} A \left((A + A^T)^2 + (A - A^T)^2 \right)$$

$$\frac{1}{2} A \left(A^2 + (A^T)^2 + 2I + A^2 (AT)^2 - 2I \right)$$

$$A \left(A^2 + (A^T)^2 \right) \quad A^T = A^{-1}$$

$$= A^3 + AA^T A^T$$

$$= A^3 + A^T$$

Question:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \beta & \alpha \\ 0 & \alpha & \beta \end{bmatrix} = A \det(2A)^3 = 2^{21}$$

Find one of value of α (or β) α, β both integers.

Options:

- (a) 3
- (b) 17
- (c) 9
- (d) 6

Solution:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \beta & \alpha \\ 0 & \alpha & \beta \end{bmatrix} = A$$

$$\det(2A)^3 = |2A|^3$$

$$= (8|A|)^3 = 2^{21}$$

$$= 2^9 \cdot |A|^3 = 2^{21}$$

$$\Rightarrow |A|^3 = 2^{12}$$

$$= \beta^2 - \alpha^2 = 2^4 = 16$$

Question:

$$f(x) = 4\sqrt{2}x^3 - 2\sqrt{2}x - 1$$

$S-1 \ f\left[\frac{1}{2}; 1\right] \rightarrow R$; $f(x)$ intersection x axis at 1 point S-2 $f(x)$ intersection x axis at

$$x = \cos \frac{\pi}{12}$$

Solution:

$$S - 1$$

$$f(x) = 4\sqrt{2}x^3 - 2\sqrt{2}x - 1$$

$$f\left(\frac{1}{2}\right) = 4\sqrt{2} \times \frac{1}{8} - 2\sqrt{2} \times \frac{1}{2} - 1 = \sqrt{2} - \sqrt{2} - 1 = -1$$

$$f(1) = 4\sqrt{2} - 2\sqrt{2} - 1$$

$$f\left(\frac{1}{2}\right) \cdot f(1) < 0$$

$$f'(x) = 12\sqrt{2}x^2 - 2\sqrt{2} = 0$$

$$= x^2 = \frac{1}{6}$$

$$x = \pm \frac{1}{\sqrt{6}} \notin \left[\frac{1}{2}, 1\right]$$

$$= 2\sqrt{2} - 1 > 0$$

$S - 1$ is true

Question: Sum of all 64 terms is 7(sum of terms at odd), find common ratio.

Options:

- (a) 6
- (b) 7
- (c)
- (d)

Solution:

$$\frac{a(r^{64} - 1)}{r - 1} = 7 \frac{ar(r^{64-1})}{r - 1} \Rightarrow r = \frac{1}{7}$$

Question: Event of tossing a dice and setting 2 in even no of throws.

Options:

- (a) $\frac{5}{11}$
- (b) $\frac{6}{11}$

Question: $(1 + y^2)(1 + \ln x) dx + x dy = 0$ Passes through $(1, 1)$ find $f(e) = \frac{\alpha \tan^{-1} \frac{3}{2}}{\beta + \tan^{-1} \frac{3}{2}}$.

Find $\alpha + 2\beta$

Question: $Z = \frac{1}{2} + 2i, |z + 1| = \alpha z + \beta(l + i)$

Find $\alpha + 2\beta$ or $2\alpha + \beta$

Solution:

$$Z = \frac{1}{2} + 2i, |z+1| = \alpha z + \beta + \beta i$$

$$Z+1 = \frac{3}{2} + 2i, \sqrt{\frac{9}{4} + 4} = \alpha \left(\frac{3}{2} + 2i \right) + \beta + \beta i$$

$$\frac{3\alpha}{2} + \beta = \frac{5}{2}, 2\alpha + \beta = 0 \Rightarrow \beta = -2\alpha$$

$$\Rightarrow 3\alpha - 4\alpha = 5 \Rightarrow \alpha = -5, \beta = -10$$

Question: (a, b) R(c, d) a, b, c, d $\in \mathbb{Z}$ ab - bd is divided by 5.

Options:

(a) S, R not T

(b) Not Transitive

Solution: Not Transitive

Question: $x^2 + y^2 = 169$, $5x - y = 13$, find area inside circle lying below the line.

Question: $4\cos\theta + 5\sin\theta = 1$, x is a solution Find $\tan x. \in \left[\frac{-\pi}{2}, \frac{\pi}{2} \right]$

Question: $4 \left[\frac{1-f^2}{1+f^2} \right] + 5 \left[\frac{2f}{1+f^2} \right] = I$ when $f = \tan \frac{x}{2}$